



The impact of the fourth regulatory determination on vulnerable populations

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Abstract In March of 2023, the first National Primary Drinking Water Standards for per- and polyfluoroalkyl substances (PFAS) were announced. The fourth Regulatory Determination that led to this development also included several other contaminants for consideration: 1,4-dioxane (dioxane), 1,2,3-trichloropropane (TCP), and strontium, which faced no determination at this time. In this study, the relative risks associated with these three contaminants and the two regulated PFAS, perfluorooctanesulfonic acid (PFOS) and perfluorooctanoic acid (PFOA), are analyzed on a subregion level, considering socioeconomic and racial factors in national exposure and risk levels for the U.S. population. Results indicate that PFOS and PFOA represent the greatest risk to the population in the subregions in which they are detected. Considering race and ethnicity, living in a majority-minority community may be a risk factor for exposure to strontium, while minority status did

not increase exposure risk for dioxane, TCP, PFOS, and PFOA. Additionally, total cancer and non-cancer relative health indicator (RHI) matrices indicate that majority-minority communities face significantly greater risks from strontium exposure. Regression models also confirm results for strontium but place the risk on racial/ethnic minority populations more specifically in regions with greater Hispanic/Latino community percentages. Finally, while greater poverty in a subregion is associated with significantly higher cancer and non-cancer RHI values for dioxane, strontium, and TCP, after controlling for state-level variations, multi-level models reveal that greater poverty is associated with significantly lower risk from these three contaminants.

Keywords UCMR 3 · Regulatory determination · Health risk · Drinking water · Socioeconomic disparity

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Introduction

Recent regulatory decisions have been made for several contaminants of emerging concern (CECs), including 1,4-dioxane (dioxane), 1,2,3-trichloropropane (TCP), strontium, and the per- and polyfluorinated substances (PFAS) perfluorooctanesulfonic acid (PFOS) and perfluorooctanoic acid (PFOA). General public awareness of PFAS has increased following the release of the film *Dark Waters* and recent news

articles referring to them as “forever chemicals” (Tingley, 2023), causing a spike in PFAS-related content on social media beginning in 2018 (Tian et al., 2022). While PFOS and PFOA are now national priorities for regulation, dioxane has been part of the discourse for over 40 years and is still unregulated at the federal level (McElroy et al., 2019). A quantitative exploration of the occurrence and health risk data supporting federal regulatory decisions is needed to evaluate the health impact of regulations, or the lack thereof, and if a segment of the U.S. population is favored by the inclusion or exclusion of certain CECs in recent regulatory decisions.

For the U.S. Environmental Protection Agency (EPA) to regulate a contaminant in drinking water, three criteria must be met under the Safe Drinking Water Act (SDWA): (1) the contaminant must have adverse health effects; (2) the contaminant must be found, or be substantially likely to be found, in public water systems with a frequency and at levels of concern; and (3) there must be a meaningful opportunity for health risk reduction, in the sole judgement of the EPA Administrator. These criteria are summarized here as adverse health effects, meaningful occurrence, and meaningful opportunity. Of the 11 contaminants discussed within the Fourth Regulatory Determination finalized in February of 2021 (U.S. Environmental Protection Agency., 2021), a positive determination was made for PFOS and PFOA. Conversely, no determination was made for three contaminants—strontium, dioxane, and TCP—as several barriers to determination were presented, including limitations in relevant adverse health effects and treatment feasibility, potential risk to public health, and analytical techniques, respectively. Six contaminants were given a negative determination and were not explored further in this study, as the analysis was centered on evaluating the health impact of positive determinations and those contaminants with limitations to determinations.

Strontium was found to occur in exceedance of the health reference level of 1500 µg/L in 5.7% of public water systems in the U.S. (Adamson et al., 2017). Adverse health effects from exposure to strontium include the weakening of bones in adults and problems with bone growth in children due to the interference with the absorption of calcium (Agency for Toxic Substances and Disease Registry, 2004). Dioxane, used as a solvent in chemical manufacturing, in

laboratories, and found in consumer personal care products, was detected above the health reference level of 0.35 µg/L in 6.9% of public water systems serving 29.4 million Americans (Adamson et al., 2017); exposure to significant amounts of dioxane can cause liver and kidney disease (Agency for Toxic Substances and Disease Registry, 2012). TCP, a by-product of chemical manufacturing, may affect the liver, kidney, respiratory tract, heart, and eyes, based on studies of both animals and humans (Agency for Toxic Substances and Disease Registry, 2021a). Severe limitations exist in the level of detection for this contaminant, leading to a lack of occurrence data within the U.S., while the health reference level from the U.S. EPA’s Third Unregulated Contaminant Monitoring Rule database (UCMR 3) is 0.0004 and 0.04 µg/L for a 10^{-6} and 10^{-4} cancer risk, respectively, and the minimum reporting level from UCMR 3 for this contaminant was 0.03 µg/L (U.S. Environmental Protection Agency Office of Water, 2017).

In March of 2023, the recommended National Primary Drinking Water Standards for PFOS and PFOA were published in the Federal Register (PFAS National Primary Drinking Water Regulation Rule-making, 2023). While these two compounds are no longer produced in the U.S., the environmental mobility and persistence of PFOA and PFOS have led to their detection in water supplies throughout the nation, with six million people consuming water with at least one PFOS or PFOA detection above the U.S. EPA’s 2016 health advisory of 70 ng/L (Hu et al., 2016).

When evaluating the meaningful occurrence criterion for regulatory determinations, the U.S. EPA primarily uses UCMR data, though several publicly available state data were used for occurrence evaluations for PFOS and PFOA. These data include details such as system size, source water type, and contaminant concentrations. However, when determining the risks to a particular community, it is well known that health outcomes are strongly correlated with many socio-economic factors such as income, access to healthcare and transportation, and even race, a concept known as the theory of fundamental causes (Link & Phelan, 1995; Phelan et al., 2010). In an evaluation of drinking water violations in the state of California, violation rates were several times higher at systems serving low-income populations and communities of color (Allaire & Acquah, 2022). This correlation

between sociodemographic factors and water quality may be due to several factors, such as low revenue or inability to pay water rates in low-income communities (Olmstead, 2004), economies of scale resulting in higher water supply unit costs for small communities (Shih et al., 2006), and structural-historical injustices (i.e., redlining, discriminatory planning policies) suppressing government support and access to social and environmental services (Glade & Ray, 2022; Schell et al., 2020). These factors also affect a community's ability to recover from hazardous events (e.g., hurricanes, mudslides, chemical spills), and as such, the Centers for Disease Control and Prevention/Agency for Toxic Substances and Disease Registry (CDC/ATSDR) developed the Social Vulnerability Index (SVI) (2021) to aid public health officials in identifying these communities, based on data collected as part of the U.S. Census.

In addition to social vulnerability, the occurrence and health risk components of U.S. EPA decision-making do not consider the relative risk between contaminants. To this end, several studies have attempted to compare contaminant risk using a health index, such as normalizing the environmental concentration to a health reference level (Rosenblum et al., 2024) or considering both cancer and non-cancer toxicity and severity to compare “cups of water” (Alfredo et al., 2017; Seidel et al., 2014). The relative health impact (RHI) equations developed in the latter studies, used in several publications to determine public health impacts (Alfredo et al., 2021; Stoiber et al., 2019), were chosen for the current study. This RHI metric is useful when comparing contaminant risks on a national scale, as authors utilized it to determine whether there was a meaningful opportunity for health risk reduction of perfluoroalkyl and polyfluoroalkyl substances (PFAS) (Alfredo et al., 2021). The metric may also be used to evaluate smaller subregions of the population, allowing for greater analysis of underlying risk factors in specific contaminant risks, such as race and poverty. Thus, the purpose of the current study is to combine the RHI and SVI matrices on a subregion-scale, consisting of national occurrence data and relevant toxicological studies for contaminants included in the latest regulatory determination, to develop an understanding of the communities in the U.S. that may be experiencing a greater risk from this selection of emerging contaminants. This analysis aims to shift the policymaking frame

in a more comprehensive and equitable direction, accounting for the risks to vulnerable communities as well as the comparative risk across contaminants.

Materials and methods

Data sources and aggregation

For this study, several publicly available databases were utilized. The first, U.S. EPA's UCMR 3, is an unregulated contaminant occurrence database that surveyed all large utilities and a subset of small utilities in the U.S. between 2013 and 2015 (U.S. Environmental Protection Agency, 2023a). This monitoring campaign required sampling to occur twice in a 12-month period for groundwater facilities (5–7 months apart) and four times in a 12-month period for surface water facilities (3 months apart). A subset of contaminants was selected due to their appearance in UCMR 3 monitoring as well as in the Regulatory Determination finalized in February of 2021 (U.S. Environmental Protection Agency., 2021) and included those given positive determinations (PFOS and PFOA) and those with no determination (dioxane, strontium, and TCP). For UCMR 3 occurrence data, 1636 subregions were represented, with these utilities serving water to 82% of the U.S. population. This dataset was utilized to assign contaminant concentrations to each public water system. The second database, the CDC/ATSDR SVI, was compiled by the agency from 2020 U.S. Census data (Agency for Toxic Substances and Disease Registry, 2021a). This dataset provided social vulnerability data at the county level. For this study, the variables of percentage per ethnic/racial minority and percentage of persons experiencing poverty were used to represent vulnerable communities, accounting for 3143 counties, municipalities, boroughs, parishes, and cities (hereafter referred to as subregions, but also known as a “statistical equivalent of a county” in U.S. Census data). Additionally, a subregion was classified as majority-minority if more than 50% of its total population was non-Hispanic White. Data from the 2020 U.S. Census shows that 12% of subregions have majority-minority populations. Poverty in the SVI database is defined by the 150% poverty line metric commonly used for benefit eligibility in the U.S., and the current study considered whether a subregion's population percent fell above or below the national average of 19.4%. For the population subset included in

this study, 20.8% of people are below the 150% poverty line metric, slightly higher than the national average.

The final dataset, the Safe Drinking Water Information System (SDWIS) Federal Reporting Services database, was used to match occurrence data from the UCMR 3 database to the census information in the SVI database (U.S. Environmental Protection Agency, 2023b), with the assumption that subregion-level census data from the SVI database was representative of the population served by the public water system within that subregion, a SDWIS and sociodemographic data aggregation method used by authors in the study of PFAS sources and detections (Liddie et al., 2023). For the SDWIS database, 3133 subregions were represented. Sources lacked sufficient sociodemographic and socioeconomic information to reliably derive effect estimates for tribal groups; therefore, tribal public water systems were excluded from the analysis.

These datasets were combined into one dataframe using R, a language and environment for statistical computing (R Core Team, 2022). For analysis, this final dataset was grouped by public water system, calculating the 95th percentile concentration per public water system identification number (PWSID), per contaminant. This resulted in 4607 unique public water systems and 1678 unique subregions with contaminant concentration and SVI data, accounting for 91% of the UCMR 3 occurrence data.

RHI calculations

Relative health index (RHI) values were calculated for each public water system, per contaminant, consistent with the methods outlined by K. Alfredo et al. (2017) and Seidel et al. (2014) resulting in a cancer RHI, non-cancer RHI, and a total RHI (RHI_{total}) value per public water system, per contaminant. The basic RHI equations include:

$$RHI_{i,total} = RHI_{i,j=Cancer} + RHI_{i,j=NonCancer} \quad (1)$$

$$RHI_{i,j} = Toxicity_{i,j} * Severity_{i,j} * Exposure_i \quad (2)$$

$$Toxicity_{i,j=NonCancer} = \frac{0.01 * I}{BW * RfD * UF} \quad (3)$$

$$Toxicity_{i,j=Cancer} = \frac{Cancer\ Slope\ Factor * I}{BW} \quad (4)$$

where I is the ingestion rate (L/day), BW is the body weight (kg), RfD is the reference dose (mg/kg-day), UF is the RfD-associated uncertainty factor, $Exposure$ is the concentration in mg/L, and $Severity$ is based on the World Health Organization global health estimates for disability weights associated with specific health outcomes. Subscript i refers to the specific contaminant and subscript j is either the cancer or non-cancer component of the risk metric.

Values used for each contaminant RHI calculation in this study are included in the Supplementary Information. Separate RHI_{total} values were calculated for various reference dose values (RfDs) for PFOS and PFOA, given the most recent toxicological data (see Table S1 in the Supplementary Information). This includes California's 2019 notification level recommendations' acceptable daily dose values for these contaminants (California Office of Environmental Health Hazard Assessment, 2019), ATSDR oral minimum risk values developed in 2021 (Agency for Toxic Substances and Disease Registry, 2021b), U.S. EPA's 2016 RfD values used in UCMR 3, and U.S. EPA's updated RfD values finalized in 2022. For additional analysis, values of RHI_{total} for the included contaminants were combined per public water system.

Odds ratios and Wilcoxon rank sum tests

For each separate contaminant and RfD considered in this study, an odds ratio was calculated to determine whether living in a majority-minority subregion may be a risk factor or a "protective" factor (not correlated with increased risk) for detecting a contaminant in a public water system in that subregion. All collected samples from UCMR 3 data for each contaminant were given a value of TRUE if the concentration was above the minimum reporting level. Rows with NA values were given values of FALSE, ensuring they indicate a non-detect for the contaminant for that sample. For SVI data for each subregion, a value of TRUE was given to subregions with a minority population percent greater than 50%, while poverty was assigned as TRUE if a subregion had poverty rates exceeding the national average of 19.4%.

For each contaminant RHI_{total} by subregion for majority-minority and majority-White subregions, a Wilcoxon test was performed to determine whether

majority-minority subregions have a statistically significant difference in median RHI_{total} values when compared to majority-White subregions.

Regression analysis

To correct for subregion-level biases due to size differences between populations served by PWSs, the mean RHI_{total} values per PWS were weighted by the proportion of the population of the given subregion that they served, shown in Eqs. 5 and 6. This assumes that the entire population of a subregion is the sum of all populations served by the PWSs in that subregion.

$$\%Population\ Served_i = \frac{PopulationServed_i}{\sum_{i=1}^n PopulationServed_i} \quad (5)$$

$$Subregion\ MeanRHI_{total}^{Weighted} = \sum_{i=1}^n (MeanRHI_{i, total} \times \%PopulationServed_i) \quad (6)$$

For regression analysis, additional subregion-level racial/ethnic composition variables were derived from 2020 U.S. Census Population Estimates and included the proportion of residents served by the PWS belonging to each racial/ethnic group. Analysis of racial/ethnic groups was restricted to subregions with at least 100 residents in the racial/ethnic group of interest, as small population numbers could violate positivity assumptions (i.e., for some subregions, the proportion of residents of a given racial/ethnic group would be zero or very near zero, and therefore, effect estimates would be extrapolated beyond the range of actual values), or bias results if small increases in the absolute number of residents resulted in large percentage differences (Petersen et al., 2012). The cutoff value of 100 residents was selected based on the assumption that most public water systems serve at least 25 people at least 60 days per year according to the EPA, with an average of four residents assumed per household (88 Stat. 1660, Safe Drinking Water Act).

RHI_{total} varied across states, though to different degrees. The intraclass correlation coefficients (ICCs)

were calculated to determine whether variations in contaminant RHI values were due to the grouping structure of either state or subregion; that is, whether differences in concentrations were due to regional factors such as terrain or climate. After results indicated high grouping variance ($ICC > 0.1$) for both state and subregion, multi-level modeling was deemed necessary to accurately determine the impact of several predictors on the RHI for a given contaminant in a public water system. The Bayesian information criterion (BIC) model comparison metric was used to select the best model to fit a subregion's mean RHI_{total} by race/ethnicity while controlling for differences in mean RHI_{total} between states, with resulting model choices listed in Table S4. This method allows for accurate modeling of the effect of variables, including race and ethnicity, on a subregion's mean RHI_{total} for each

contaminant. Measures of poverty were included in regression models after tests for collinearity with race/ethnicity indicating weak or no correlation (Pearson correlation coefficients = -0.043 – 0.130). Calculated ICCs for sum RHI_{total} per subregion indicated multi-level modeling was not necessary, shown for each RfD in Table S4; thus, standard linear regression was used for this analysis.

Results and discussion

We present our analysis in two ways. First, we consider the traditional method of evaluating exposure to varying contaminants by using the number of detections above an established health reference level to evaluate if sub-populations are at greater risk of exposure. In our second analysis, we move beyond this status quo analysis to evaluate the relative health risk of each contaminant on a normalized RHI scale and the impact on socially vulnerable populations.

Table 1 Odds ratios representing odds of exposure to contaminants among minority-majority communities and of exposure to contaminants among communities above the national average 150% poverty threshold.^a

Contaminant	OR minority- majority	<i>p</i> -value	OR poverty threshold	<i>p</i> -value
1,2,3-trichloro- propane	0.17	0.016	0.11	<0.001
1,4-dioxane	0.72	<0.001	0.75	<0.001
Strontium	2.19	0.049	1.34	0.23
PFOS	0.56	0.016	0.61	0.002
PFOA	0.61	0.006	0.39	<0.001

^aMore information regarding the OR statistics may be found in the Supplementary Information

Conventional assessment: odds ratios for contaminant detections

Pairing contaminant detections with subregion-level data designating the subregion as majority-minority or majority-White, the odds ratio (OR) was performed, shown in Table 1. From the UCMR 3 data, strontium had the most detections nationwide, with 41,906 samples above the UCMR 3 detection limit. Dioxane had the next highest number of detections with 2598 samples above the detection limit, while PFOS, PFOA, and TCP had 166, 277, and 49 detections, respectively. Out of the 1678 subregions analyzed, 388 (23%) are majority-minority. This percentage is slightly higher than data reported by the 2020 U.S. Census, where only 12% of subregions across the entire U.S. are majority-minority. This suggests that minority populations are somewhat more concentrated in subregions served by large utilities. Results from odds ratios show people living in a majority-minority community have two times greater odds of having a detection of strontium in their public water system, compared to majority-White communities ($p < 0.05$); this suggests that living in a majority-minority community may be a risk factor for exposure to strontium. The remaining contaminants included in this study had odds ratio values less than one, suggesting that living in a majority-minority community does not increase one's probability of exposure to contaminant detections in drinking water. It should be noted that the odds ratio is not equivalent to a risk ratio (or the probability of exposure), which is a common misinterpretation of ORs (Persoskie & Ferrer,

2017). The OR of 0.56 for PFOS in majority-minority subregions, for example, does not indicate that minority populations are about half as likely to be exposed to PFOS, but that the odds of exposure are lower.

Odds ratios performed with subregion-level detection data and subregions with the majority (> 50%) under the 150% poverty were not statistically significant, as only 0.9% of U.S. subregions had this level of poverty. Instead, we targeted subregion populations that have greater amounts of poverty than the national average, performing the odds ratios for subregion populations below the 150% poverty line that were above the national average of 19.4%, based on the 2020 U.S. Census. From this analysis, shown in Table 1, it was found that for dioxane, TCP, and PFOA, living in a subregion with greater poverty than the national average was not a risk factor for detecting these specific contaminants in the drinking water, with odds ratios of 0.75, 0.11, and 0.39, respectively ($p < 0.001$). PFOS also did not have an increased exposure risk in areas with greater poverty, but with less statistical significance ($p < 0.01$). The odds ratio for strontium detections was not statistically significant. These contaminants follow trends of the percent of a population in poverty being inversely associated with the water contaminant seen in other contaminant comparisons such as nitrate (Schaidter et al., 2019).

Of the contaminants included in this study, strontium was the only one with greater odds of affecting minority communities. When we focus on only detections, however, we ignore compounding health impacts. Additionally, other inequities may exacerbate the issue of strontium disproportionately affecting communities with majority-minority populations. Exposure to strontium in the diet or drinking water may lead to problems with bone growth, particularly for children with low intake of calcium (Agency for Toxic Substances and Disease Registry, 2004). A study estimating the contribution of dietary supplements to nutrition within racial subgroups found that without supplementation, 56.3% of non-Hispanic Black and 61.4% of non-Hispanic Asian adults have calcium intake levels below average intake requirements, compared to 35.9% of non-Hispanic White adults. Hispanic adults' calcium intake was closer to non-Hispanic Whites at 36.4% (Blumberg et al., 2017). Non-White Americans are also more likely to live in "food swamps,"

characterized by a high density of fast-food establishments and less access to nutritionally dense foods (Block et al., 2004; Cooksey-Stowers et al., 2017), and face greater instances of health conditions such as diabetes and hypertension (Gaskin et al., 2014; Zhang & Moran, 2017). This analysis does not consider the other water contaminants a community might be exposed to such as copper, mercury, lead, and nitrate which often impact non-White communities (Neville et al., 2022; Schaidler et al., 2019; Vanderslice, 2011).

Reframing contaminant exposure analyses: RHI_{total} by contaminant

To move beyond detection analysis, the RHI was calculated for each subregion and for each separate contaminant, allowing for a comparison of public health impacts. Once the cancer and non-cancer RHI values were calculated and summed, this RHI_{total} value was combined with sociodemographic data per subregion as well as per public water system. As shown in Fig. 1, the relative health risk of contaminants that were recently considered for primary drinking water standards varies with concentration and given

toxicological data. Focusing on the RHI values from UCMR 3 RfDs, TCP was found to be the greatest health risk for subregions in which it was detected, despite the fact that the number of detections ($n=49$) were the lowest in this study. Evaluating a contaminant without this larger context could skew regulatory decisions. A well-known issue with TCP occurrence data is the relatively high detection limit due to analytical method limitations. In UCMR 3, the minimum reporting level was 0.03 $\mu\text{g/L}$, while the health reference level was 0.0004 $\mu\text{g/L}$. It is highly possible that more communities are exposed to TCP at these lower concentrations. Despite the high relative risk, a federal regulatory decision was deferred until national occurrence at concentrations relevant to public health could be accounted for. Note that PFOS and PFOA HRLs and RfDs have varied depending on the assessment, which impacts our calculations of relative risk. Once recent RfDs are considered in the RHI calculations, however, the risks of PFOS and PFOA in drinking water increase substantially, particularly for the most recent U.S. EPA RfDs released in 2022. This means that, for subregions with PFOS or PFOA detections, these contaminants represent the greatest

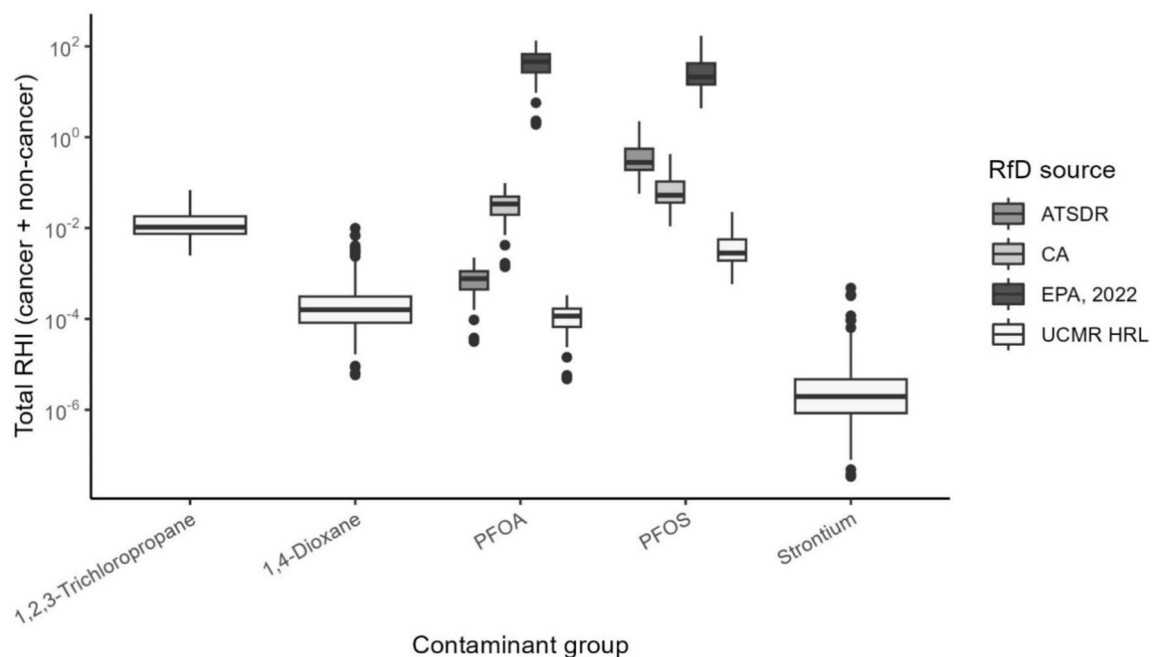


Fig. 1 Subregion-level RHI_{total} values per contaminant. The box represents the 25th percentile and the 75th percentile of the data, while the horizontal line inside the box represents the median

health risk out of this subset of contaminants when updated RfD values are considered.

Considering each contaminant's RHI_{total} by sub-region for majority-minority and majority-White subregions, a Wilcoxon test was performed to determine whether majority-minority subregions have a statistically significant difference in median RHI_{total} values when compared to majority-White subregions. Strontium results indicate that there is a significant difference for this contaminant: majority-minority subregions are associated with a higher non-cancer RHI for strontium ($p < 0.05$, Figure S3). This is consistent with the detection results from the odds ratio for strontium. Similarly, there was no significant difference between median RHI_{total} values between these populations for the other contaminants in this study. When comparing RHI_{total} values between subregions above the national average poverty metric of 19.4% to those at or below this threshold, a significant difference was found between the medians of these

groups for dioxane (Figure S4), strontium, and TCP ($p < 0.05$), while no significant difference was found for PFOS and PFOA. This indicates that communities with higher poverty rates may also be exposed to higher concentrations of—and thus greater risk levels for—dioxane, strontium, and TCP. While PFOS and PFOA have the highest relative risk based on RfD and health and/or regulatory thresholds, they might not pose the greatest threat to subpopulations in the U.S. Strontium repeatedly had the highest detections and, when the concentration of exposure is considered, the highest RHI_{total} in majority-minority communities, and dioxane, TCP, and strontium had a higher correlation between high calculated RHI and above-average poverty rates.

To determine the impact of sociodemographic factors on individual contaminant RHI_{total} values, multi-level modeling was performed, with model results shown in Table 2. While model 1 selected predictors using the BIC method, model 2 included

Table 2 Multilevel model coefficients and significance^a

Contaminant	RfD source	Model ^b	Percent Hispanic	Percent minority	Percent poverty	Percent Non-White Other	Percent Black
TCP	EPA	Model 1	9.24E-06 ***				
		Model 2	1.07E-05 ***				
Dioxane	EPA	Model 1		2.91E-06 ***	− 1.18E-05 ***	− 4.51E-06 NS	3.81E-06 NS
		Model 2	3.07E-06 ***		− 4.75E-06 ***	3.69E-06 **	2.46E-06 ***
PFOA	ATSDR	Model 1				1.25E-06 ***	
		Model 2	3.80E-07 NS		− 6.09E-07 NS	1.21E-06 *	− 3.87E-07 NS
	CA	Model 1				5.50E-05 ***	
		Model 2	1.67E-05 NS		− 2.68E-05 NS	5.34E-05 *	− 1.70E-05 NS
	EPA	Model 1				1.88E-07 ***	
		Model 2	5.69E-08 NS		− 9.13E-08 NS	1.82E-07 *	− 5.80E-08 NS
	EPA, 2022	Model 1				7.50E-02 ***	
		Model 2	2.28E-02 NS		− 3.65E-02 NS	7.29E-02 *	− 2.32E-02 NS
PFOS	ATSDR	Model 1		1.45E-04 ***			
		Model 2	1.55E-04 *		− 1.28E-04 NS	4.10E-04 **	1.08E-04 NS
	CA	Model 1		2.77E-05 ***			
		Model 2	2.94E-05 *		− 2.44E-05 NS	7.79E-05 **	2.05E-05 NS
	EPA	Model 1		1.47E-06 ***			
		Model 2	1.56E-06 *		− 1.30E-06 NS	4.15E-06 **	1.09E-06 NS
	EPA, 2022	Model 1		1.10E-02 ***			
		Model 2	1.17E-02 *		− 9.74E-03 NS	3.11E-02 **	8.20E-03 NS
Strontium	EPA	Model 1	1.56E-07 ***	− 6.44E-08 ***	− 1.58E-07 ***		
		Model 2	9.19E-08 ***		− 1.69E-07 ***	− 9.90E-08 **	− 5.30E-08 **

^a $p < 0.05$, $**p < 0.01$, $***p < 0.001$; NS, not significant; ^bmodel 1 predictors were selected by BIC, model 2 included the same predictors for all contaminants

the same predictors for all contaminants. Results from both model 1 and model 2 for TCP suggest that the percentage of a subregion that is Hispanic/Latino is associated with a significant increase in RHI_{total} for this contaminant ($p < 0.001$), while greater percentages of poverty are associated with a decrease in TCP's RHI_{total} . For 1,4-dioxane, percentages of poverty for both model 1 and model 2 were significantly associated with a decrease in RHI_{total} ($p < 0.001$), while minority population percentages from model 1 and Non-Hispanic Black, Hispanic/Latino, and Non-White Other percentages from both models were significantly associated with increases in RHI_{total} ($p < 0.001$). Results for PFOA suggest that increased percentages of a subregion population that is Non-White Other may significantly increase the RHI_{total} values associated with this contaminant, with varying degrees of magnitude dependent on the value of the RfD, as expected (model 1, $p < 0.001$; model 2, $p < 0.05$). The magnitude of the PFOS RHI_{total} , on the other hand, was significantly associated with increased percentages of minority populations for model 1 ($p < 0.001$) and Hispanic ($p < 0.05$) and Non-White Other ($p < 0.01$) for model 2. Results from both models 1 and 2 indicate that greater poverty and Hispanic population percentages are associated with significantly lower and higher RHI_{total} values for strontium, respectively ($p < 0.001$). Additionally, greater minority, Non-White Other, and Black population percentages are associated with significantly lower strontium RHI_{total} values.

A major assumption in the RHI framework is that contaminant RHIs are additive (Alfredo et al., 2017, 2021; Seidel et al., 2014). This cumulative RHI metric clarifies a subregion's overall risk from any number of contaminants with available toxicological and occurrence data. When comparing subregions' sum

of RHI_{total} values for the CECs included in this study to the percent minority population, no correlation was found (see Supplementary Information, Figure S5). Similarly, no significant difference was found between the median sum of the RHI_{total} values for majority-White subregions and those of majority-minority subregions, Figure S6, suggesting that the summed overall risk from these contaminants at the subregion level does not disproportionately affect minority communities. No correlation was found between the sum of contaminant RHI_{total} values and the percent of the population below the 150% poverty line. However, a significant difference was found between the median sum of the RHI_{total} values for subregions above the 19.4% national average poverty metric and those at or below this national average ($p < 0.05$, Figure S7). The higher median sum RHI_{total} suggests that communities experiencing greater rates of poverty may not be at greater risk from the subset of contaminants considered for regulatory determination.

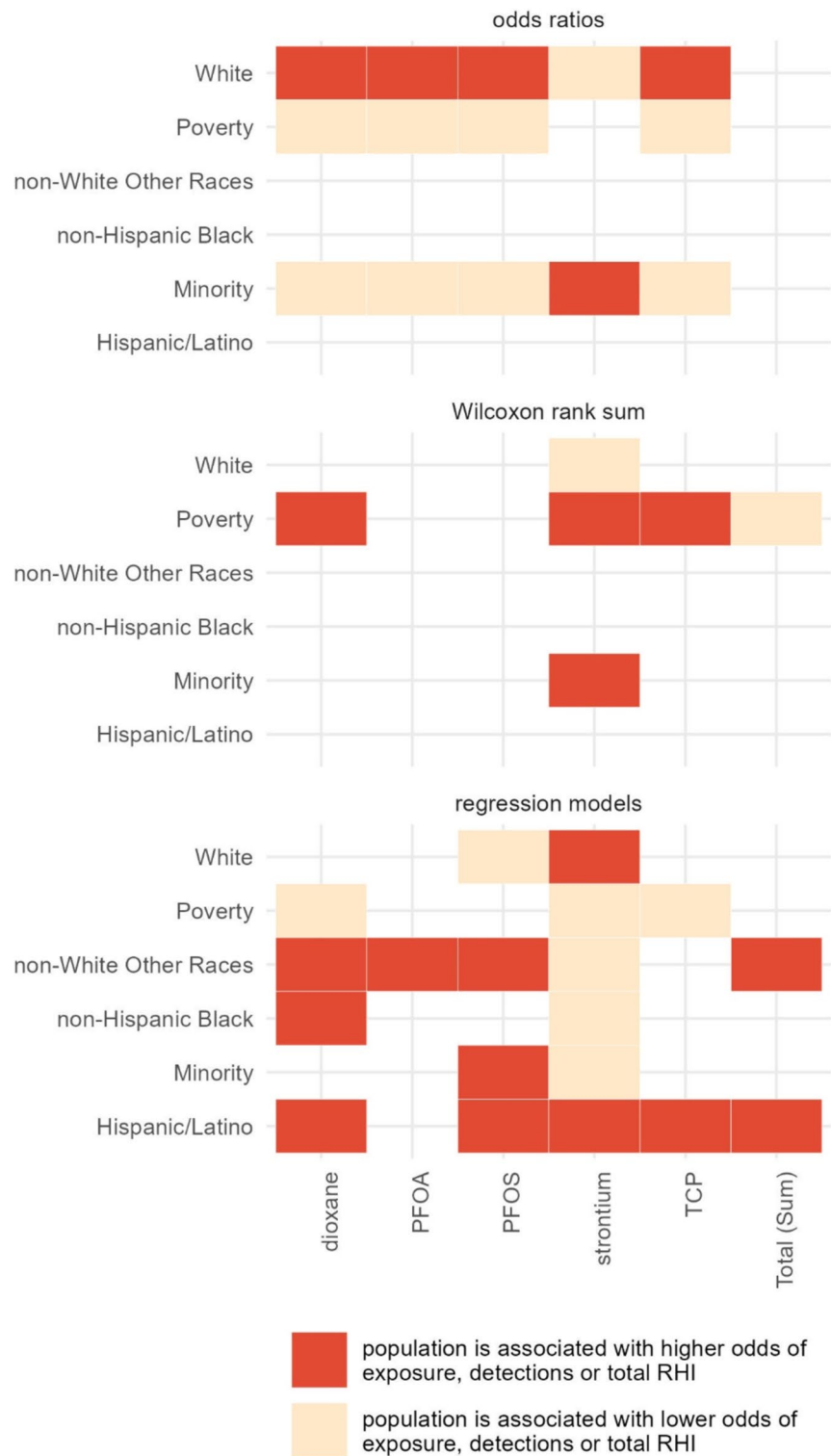
Considering the results from linear regression, the overall risk level presented in Table 3 indicates that the RfD used to calculate PFOS and PFOA risk greatly impacts the calculated risk for the populations considered. The regression results for sum RHI_{total} confirm those discussed previously in the multilevel models for individual contaminants, as a higher PFOS and PFOA RfD from UCMR 3 would mean that TCP, dioxane, and strontium carry greater weight in the overall sum RHI_{total} values, resulting in a significant increase in risk for subregions with increased Hispanic population percentages ($p < 0.001$) for the EPA RfD. Significant increases in RHI_{total} values were also found in subregions with greater population percentages of Non-White Other ($p < 0.05$) for models 1 and 2 when the ATSDR RfD was considered in RHI calculations.

Table 3 Sum RHI_{total} regression model coefficients and significance^a

RfD source	Model ^b	Percent Hispanic	Percent poverty	Percent Non-White Other	Percent Black
ATSDR	Model 1			4.28E-04	*
	Model 2	-4.27E-05	NS	-1.37E-04	NS
CA	Model 1			1.16E-04	NS
	Model 2	-5.78E-07	NS	-4.24E-05	NS
EPA	Model 1	1.07E-05	***		
	Model 2	1.07E-05	***	-3.05E-06	NS
EPA, 2022	Model 1			6.68E-02	NS
	Model 2	-6.25E-03	NS	-3.09E-02	NS

^a* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; NS, not significant; ^bmodel 1 predictors were selected by BIC, model 2 included the same predictors for all contaminants

Fig. 2 Summary of significant results for each analysis of RHI_{total} ($p < 0.05$). Note: all p -values in this figure were $p < 0.05$, with some values < 0.001 , indicating high statistical significance



Considering the results from all methods used in this study to evaluate populations at greater risk, including odds ratio, Wilcoxon rank sum test, and regression models, a few populations are consistently affected by this group of contaminants, as shown in Fig. 2. Risk for each population varies substantially depending on the evaluation; for instance, strontium is shown to impact minority populations according to odds ratio and Wilcoxon results, but once state-level effects are considered, greater minority population percentages are associated with lower strontium RHI_{total} . Notably, all contaminants except for PFOA are associated with greater risk levels in Non-White Hispanic/Latino populations according to regression models.

Our results overall show that, unexpectedly, increased poverty is not significantly associated with negative water quality outcomes with regard to this subset of contaminants. One possible explanation for this is the focus of UCMR 3 monitoring data on large utilities (serving > 10,000 people). In a study of water system violations, race/ethnicity, and socioeconomic status, McDonald and Jones (2018) found that increasing the stratification of water utility sizes reveals differences between violation rates in vulnerable communities, including significantly increased odds of repeat violations corresponding to increases in uninsured residents for all systems except those serving $\geq 50,001$ people. Additionally, increases in household median income also increased the odds of repeat violations, but only in medium-sized utilities (which they define as 10,001–50,000 people). These results may indicate that the bias in UCMR 3 data towards large utilities could be underrepresenting the impacts of these contaminants on communities experiencing greater poverty, if many are in smaller water systems that are more likely to have repeat violations.

Conclusions

Recent justifications by the U.S. EPA for declining to make determinations for several emerging contaminants are published in the Federal Register, including a gap between what is measurable and what is relevant to health (TCP) and a need to further understand potential health effects and national occurrence (dioxane, strontium). Meanwhile, the agency has proceeded with regulating PFOS and PFOA, stating that UCMR 3 occurrence data and recent toxicological studies, along

with the availability of treatment technologies, meet the three requirements under the SDWA of establishing a National Primary Drinking Water Regulation. We present a different framework for comparing contaminants being considered for regulation to ensure public health is being protected for all populations. These risk evaluation methods are intended to be used in evaluating the potential benefits of national drinking water regulations—that is, *will the most vulnerable populations benefit from a regulation?* If a regulatory decision was made based on the nationwide frequency of detections above a reference dose, then a regulation for strontium would be the most beneficial to minority communities, as they have greater odds of detecting this contaminant. Indeed, according to the RHI_{total} values calculated here, greater relative risk is associated with strontium in minority communities, compared to majority White communities. However, this work shows that state-level effects are significant in the evaluation of relative risk to these subpopulations. Given this, national regulations for dioxane and PFOS would have the greatest impact on addressing risk to vulnerable populations in the U.S., as three out of the five vulnerable groups are considered to face significantly higher risk from these contaminants.

This work demonstrates a regulatory decision-making method that expands on the current policy of evaluating the national occurrence of a contaminant—whether it is detected in enough public water systems at concerning concentrations—and considers the cumulative health risk of exposure as well as a comparative assessment between other contaminants. This goes beyond contaminant detections in a way that enables a discussion of contaminant prioritization, not only for contaminants facing regulatory determinations but also for regulated contaminants that may represent an even greater health risk for certain subregions of the nation. Overall, these results confirm that multilevel modeling is necessary to adjust contaminant risk, as these sociodemographic factors vary by state and could impact the interpretation of occurrence data. Furthermore, contaminant detections do not incorporate relative risk, which may be addressed with the RHI metric to facilitate contaminant risk comparisons grounded in both measured concentrations and recent toxicological evaluations.

Additionally, when developing drinking water regulations based on national occurrence data and contaminants' health impacts, consideration for the socioeconomic vulnerability of populations should be included; this goes beyond the traditional vulnerable groups

that may be considered in toxicological data, such as infants and those who are pregnant or breastfeeding. Populations living below the poverty line and racial or ethnic minority communities face greater impacts from health conditions such as diabetes and hypertension and experience increased nutritional deficiencies. Combining social vulnerability data with a relative health metric such as the one used in this study helps to develop a more detailed picture of what the greatest drinking water risks are to the most vulnerable communities in the U.S., which could in turn inform water quality interventions beyond regulatory measures, such as distributing point-of-use filters, providing grants for infrastructure improvements, or increasing sample collection to identify contaminant hotspots. We must also emphasize that, while this study identifies the contaminants with the greatest relative risk to vulnerable populations, exposure to the remainder of contaminants may also pose significant health risks and thus should not be excluded from future regulatory discussions.

This study has potential limitations due to the use of UCMR 3 data, which presents contaminant occurrence data collected from primarily large utilities serving 82% of the U.S. population. This dataset does not perfectly reflect the overall national sociodemographic data's subgroup distributions (e.g., 12% of subregions in the nation are majority-minority, while 23% of subregions in the UCMR 3 dataset are majority-minority). Additionally, severe limitations exist in the data for TCP, in particular, due to the relatively high detection limit for this contaminant during the UCMR 3 monitoring campaign. However, as this data was used in developing the EPA's decisions, we feel that this data represents the information available to regulators at the time of the fourth Regulatory Determination and, as such, allows for a comparison between the risk evaluations presented in this work and those presented in the documentation of the Regulatory Determination.

Authors' contribution Andrew Black jointly conceived the study with Katherine Alfredo; A.B. cleaned and assembled input data, wrote code to calculate and evaluate contaminant risk, analyzed code output, and wrote the manuscript. Natchaya Luangphairin designed and implemented the regression models, wrote model code, and analyzed model output. K.A. supervised research activity, planning, and execution. All authors discussed results and implications and edited the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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